TEVIX

Team Name - Sentinel

Presenter Names

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TEVIX - Technology guided by ecology, driven by intelligence, creating value with exponential impact.

The Perishable Protein Challenge

Biological Spoilage

Protein-rich foods (shrimp, fish, meat, dairy, eggs) possess exceptionally high moisture and nutrient levels.

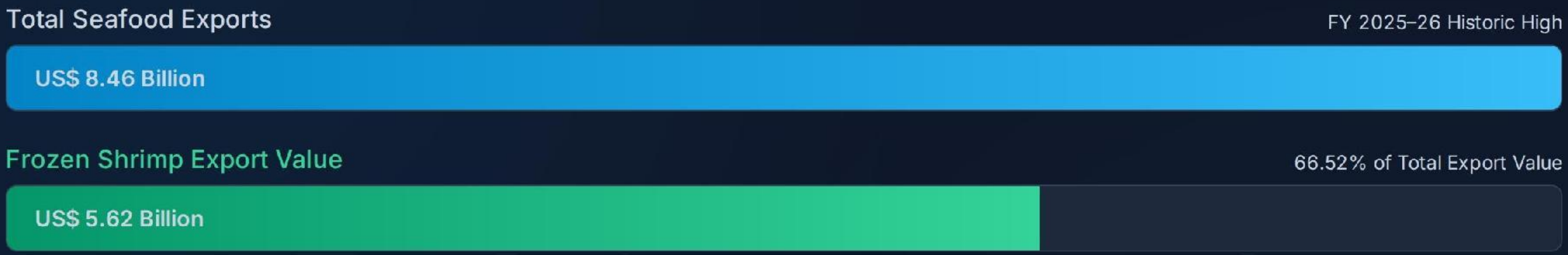
This biological makeup creates a high susceptibility for rapid microbial growth and rapid quality decay.

Unpredictable Cold Chains

Standard refrigeration slows decay, but sudden, unmonitored temperature breaches still cause unexpected spoilage.

Conventional packaging merely physically protects the product—it cannot monitor or communicate active freshness.

India's Seafood Export Dominance



Targeting major high-compliance global markets including the **United States** and **China** where traceability standards are strict.

Huge premium opportunity for **SensoPack** to mitigate cold-chain damage and solidify export quality in a multi-billion dollar segment.

Large Market Opportunity

This creates a significant opportunity for **SensoPack**, an intelligent packaging solution that provides real-time freshness monitoring, reduces spoilage, improves export quality, and enhances cold-chain traceability.

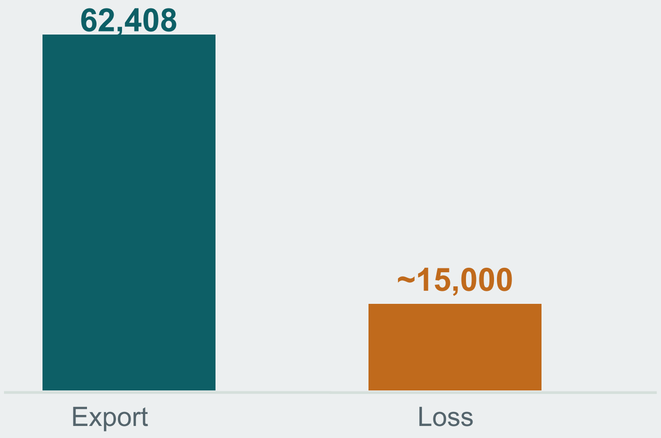
PROBLEM SIGNIFICANCE

India exports the world's shrimp — and loses a fortune to spoilage

Farm / Landing	Cold store / Transit	Port / Export	Retail / Consumer
melanosis	thermal excursion	quality downgrade	

Shrimp spoils in 3–14 days by temperature — a static "Best By" label is blind to it

INDIA SHRIMP EXPORT vs LOSS (₹ cr)



FY 2024–25 · 1.69M MT exported

69%
of India's seafood dollar earnings come from shrimp

$2.71B
India → USA shrimp alone (FY25) — the premium market

~35%
of harvested fish/seafood lost globally before a plate

2 gaps
no active protection; no real-time information

Unmet Informal Sector Needs

The Active Tag Cost Barrier

Unviable IoT Pricing: Standard active electronic tracking tags cost over $2.50 per unit, creating a major financial barrier for retail implementation.

Strict Margin Caps: Safe integration requires the monitoring sensor to fit strictly between 5% and 8% of the package's total value.

Domestic Supply Chain Realities

90% Unrefrigerated Logistics: The vast majority of the domestic Indian food sector operates completely free of continuous refrigeration systems.

Deficient Infrastructure: Missing temperature checkpoints make point-of-sale visual verification completely unreliable for real safety control.

Creative & Original Solution

Dual-Function Smart Bioactive Film

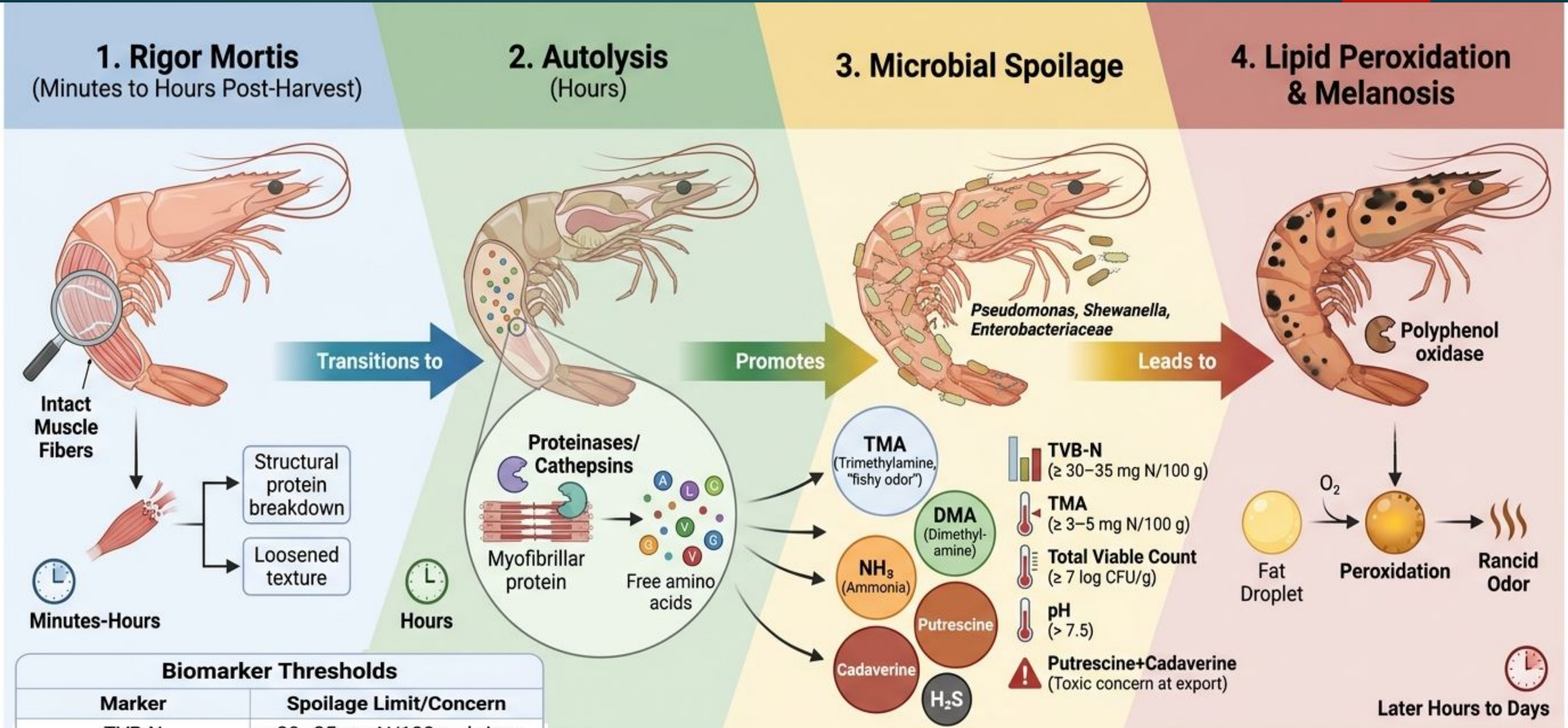
Combines active **antimicrobial protection** and **real-time freshness sensing** in a single biodegradable film. Unlike standard packs, it preserves food and tracks its biological quality simultaneously.

Circular Bioeconomy Modular Design

Transforms raw **shrimp shell waste** into a value-added chitosan biofilm integrated with natural essential oils. Includes a **reusable NFC electronics tag** to drastically reduce plastic and electronic waste.

Multi-Tier Democratic Platform

Scales from **battery-free color-changing indicators** for local micro-vendors to advanced **AI-enabled NFC smart sensing** for international exporters, bringing intelligence to all supply chain tiers.



Biomarker Thresholds

Marker	Spoilage Limit/Concern
TVB-N	≥ 30–35 mg N/100 g shrimp
TMA	≥ 3–5 mg N/100 g (fishy odor)
Total Viable Count	≥ 7 log CFU/g
pH	> 7.5
Putrescine+Cadaverine	Toxic concern at export

- Shrimp spoilage is not a single event — it is a cascade of four overlapping biochemical and microbial processes after death
- These biomarkers can be the sensing targets.
- Any embedded sensor system must track one or more of these molecular signatures in real time.

Ref: Das, J., &

Technical Feasibility and Scientific Soundness

Multi-Biomarker Detection

Scientifically validated parameters monitor product degradation securely across the complete food spoilage process:

Trimethylamine (TMA) Monitoring

Captures localized volatile amines indicating protein decay, particularly in fish and poultry.

Total Volatile Nitrogen (TVB-N)

Evaluates broader nitrogenous compounds to give a complete freshness and shelf-life snapshot.

Real-Time pH Shift Tracking

Detects minute electrochemical shifts dynamically within the internal microenvironment.

Technical Feasibility and Scientific Soundness

Advanced Sensing Platform

MIP Sensors

Integrates Molecularly Imprinted Polymer (MIP) electrochemical receptors. They function like synthetic antibodies, offering absolute selectivity for target degradation compounds.

CMOS ASIC

Features an ultra-low-power 130 nm CMOS application-specific integrated circuit (ASIC). Converts unstable raw chemical signals into clean, stable digital freshness data.

NFC Logistics

Powers passive, battery-free wireless reads. Delivers real-time freshness scans, automatic shelf-life estimations, and end-to-end AI cold-chain tracking.

Technical Feasibility and Scientific Soundness

Manufacturing Scale

High-Volume Output: Manufactured on rapid, high-throughput assembly lines using slot-die coating and roll-to-roll printed electronics.

Ultra-Thin Profile: Film layers measure <20 μm, enabling highly flexible integration with flexible packaging substrates.

Seamless Compatibility: Adapts directly to existing hot-jaw heat sealers and standard processing machinery. Zero factory overhaul required.


SENSOPACK™ – SMART, SUSTAINABLE & INTELLIGENT SHRIMP PACKAGING
Four-Layer Modular Architecture for Real-Time Freshness Monitoring and Cloud Intelligence
LAYER 1 – ACTIVE BIOFUNCTIONAL PACKAGING
Protects & Extends Shelf Life
Circular Chitosan Film (From Shrimp Shell Waste)
Essential Oils (Thymol, Carvacrol, Clove Oil)
Barrier Properties Low OTR, Balanced WVTR
Inhibits Bacteria • Reduces Oxidation • Extends Shelf Life
LAYER 2 – DISPOSABLE SMART SENSOR FILM
Detects Spoilage Markers
Spoilage Gases TMA, NH_3, TVB-N
Anthocyanin Indicator (Color Change)
MIP Sensor (Selective TMA Detection)
Electrical Signal Generated
Purple
Green
Brown
Completely Disposable • Food Contact Safe • Biodegradable
LAYER 3 – REUSABLE SMART ELECTRONIC TAG
Processes, Stores & Transmits Data
Signal Processing
Ultra Low Power ASIC
Memory (Data Storage)
Temperature Logger
((NFC))
NFC Antenna (13.56 MHz)
Clip-On Thin Tag • Reusable (100–500 Cycles) • Replaceable Battery
LAYER 4 – CLOUD AI & DIGITAL TWIN PLATFORM
Predicts Shelf Life & Ensures Export Quality
Scan with Smartphone / Retail Reader
Cloud Database (Secure)
AI Prediction Engine (Kinetics + ML Models)
Freshness Score & Expiry
Shelf Life Prediction • Cold Chain Monitoring • Export Assurance
TOTAL THICKNESS < 20 µm (Film)
TAG THICKNESS ~ 2 mm
LIGHTWEIGHT • FLEXIBLE • COMPACT
FOOD CONTACT SAFE
< 0.4 mm
SensoPack™
SMART PACKAGING
SMART FUTURE
FRESH
SAFE
SUSTAINABLE
NFC
REUSABLE NFC TAG (CLIP-ON)
60 mm
15 mm
2 mm
Ultra Low Power ASIC
NFC Type 2
Temperature Sensor
Data Logger
Replaceable Battery
Water Resistant
Reusable 100–500 Times
DISPOSABLE SMART SENSOR FILM
(Integrated in Strip)
Layer 1
Layer 2
Top Protective Layer (Moisture & Oxygen Barrier)
Active Bio Layer (Chitosan + Essential Oils)
Colorimetric Indicator (Anthocyanin)
MIP Sensor Layer (TMA / Biogenic Amines)
Base Sealant Layer (Food Contact Safe)
VISUAL FRESHNESS INDICATION
Anthocyanin Color Change
Fresh (Purple)
Good (Green)
Caution (Olive/Brown)
Spoiled (Brown)
Fresh Shrimp Packed
Active Film Protects & Monitors
Spoilage Gases Generated
SensoPack™
Sensor Film Detects & Generates Signal
Tag Processes & Stores Data
NFC Scan (Instant)
Data to Cloud (Secure)
AI Predicts Shelf Life & Quality
SAFE / CAUTION / UNSAFE Dynamic Expiry
SUSTAINABLE END-OF-LIFE CYCLE
1 After Use (Shrimp Consumed)
2 Disposable Film Biodegrades (Compostable)
3 Remove Tag from Package
4 Cleaning & Testing
5 Recharge / Replace Battery (If Needed)
7 Attach to New Package
Reuse Again (100–500 Cycles)

Impact Potential - Scale, Depth, and Reach

Enhancing Food Safety & Health

HEALTH FIRST

- **Real-Time Biomarker Tracking:** Enables continuous detection of volatile shrimp spoilage markers including TMA and TVB-N.
- **Public Health Preservation:** Active prevention of toxic seafood consumption associated with biogenic amines.
- **Unsafe Practice Deterrence:** Direct reduction in reliance on toxic chemical preservatives like formalin.

Impact Potential - Scale, Depth, and Reach

Reducing Supply Chain Loss

FINANCIAL IMPACT

Mitigating high value seafood waste through active tracking.

Distress Pricing Mitigation: Eliminates standard 20% to 25% distress markdowns through validated freshness metrics.

Protecting Stakeholder Margins: Maximizes revenue across the seafood value chain for fishermen, exporters, and retailers.

Source: image_2af393.jpg

Affordable & Scalable Solutions

Three-Tier Model

Robust and tiered packaging structures pricing to ensure high availability across tiered global operations.

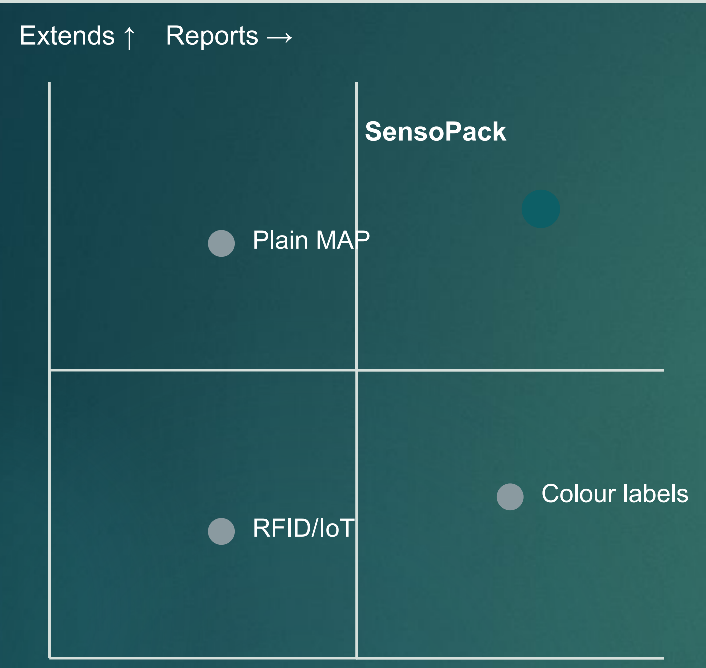
Rural to Export

Designed for universal feasibility, validating freshness at rural fish landing harbors to elite global container chains.

Macro-Industry Lift

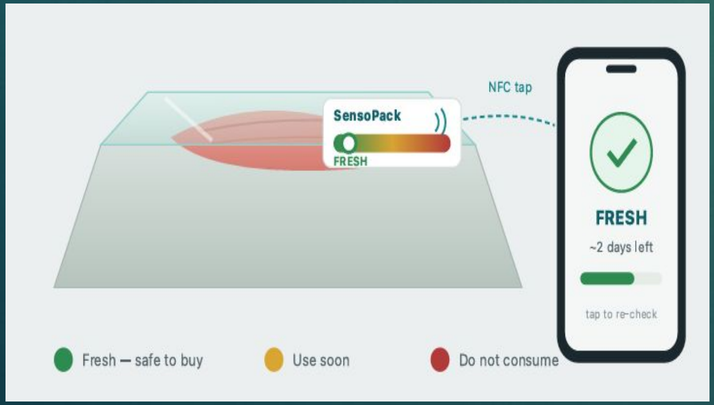
Strengthens India's multi-billion dollar export industry through reliable safety certifications and compliance tracking.

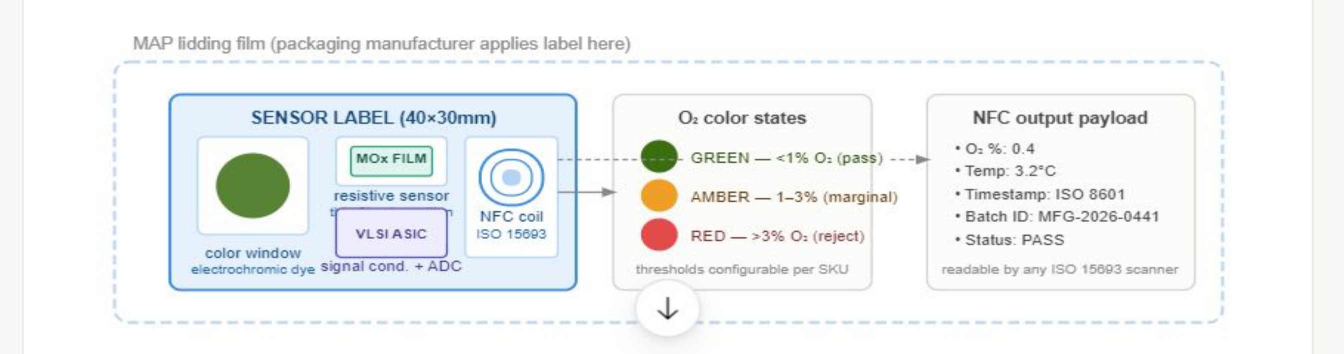
The only solution that both protects and reports — affordably



Capability	SensoPack	Colour labels Senoptica	IoT tags TrackingPacking	Add-on Blakbear	Plain MAP
Extends shelf life	Yes	No	No	No	Partial
Reports freshness	Yes	Yes	Temp only	Yes	No
Naked-eye / no scanner	Yes	Needs scanner	Phone/reader	Reader	—
Packaging-integrated	Yes	Yes	No	Bolt-on	Yes
Cost / unit	<₹5–120	>High	>$2.50	Med	Low
Biodegradable	Yes	Varies	No	No	No

Every incumbent solves one axis. SensoPack is the only entry combining active protection, honest sensing, biodegradability and affordability in one package.





Prototype Quality and Stage of Development

Lab-Scale Proof of Concept

Successful Seafood Formulation

Developed a robust, highly transparent, biodegradable chitosan-based smart packaging film tailored strictly to handle delicate seafood applications.

Integrated Material Platform

Proven synergy combining antimicrobial compounds into a biodegradable matrix. This acts as the key foundational layer for integrating future digital smart-sensing electronics.

Prototype Specifications & Hurdles

Parameter	Prototype Current Status	Scale-Up Hurdles
Transparency	High optical clarity; zero internal phase separation.	⚠️ Hold on thin-gauge industrial rolls
Integrity	Extremely clean edges, highly uniform, no drying cracks.	⚠️ Survive heat-sealing and MAP constraints
Resilience	Forms a highly durable standalone protective sheet.	⚠️ Maintain high moisture barrier in cold storage

Next Development Milestones

Smart Hardware

Integrating freshness sensing nodes and NFC-enabled electronics directly inside the barrier films.

AI Shelf-Life Analytics

Building predictive shelf-life modeling tools using advanced sensor telemetry and chemical analysis.

Pilot Validation

Deploying real-world validation trials using refrigerated shrimp within standard supply storage conditions.

Viability of the Business

Scalable & Cost-Effective Model

Low-Cost Innovation: Designed as an affordable, high-volume smart packaging solution utilizing biodegradable materials and printed electronics.

Modular Architecture: Seamless design structure enables highly efficient and inexpensive manufacturing scale-up.

Large-Scale Footprint: Engineered to match and support immediate deployment requirements across domestic and global seafood supply chains.

Viability of the Business

Multiple Revenue Streams

Packaging Films

High-volume B2B enterprise sales of eco-friendly, sensor-embedded protective smart films to food processors and exporters.

Premium NFC Tags

Monetization of reusable NFC-enabled tags tracking critical telemetry for cold-chain operators and high-value logistics.

SaaS AI Platform

High-margin recurring subscription revenue through our predictive Freshness Intelligence AI analytics portal.

Viability of the Business

Market Adoption & Value Proposition

Core Value Proposition

SensoPack seamlessly fuses highly affordable smart packaging, state-of-the-art reusable electronics, and cloud-based AI analytics into one scalable ecosystem that drastically reduces global food waste while boosting logistics margins.

Compatibility

Works with existing MAP packaging lines

Product Range

Tiered visual to AI indicators

Implementation Model

KEY PARTNERS

- Shrimp processors and exporters
- Packaging manufacturers
- Chitosan suppliers
- Essential oil manufacturers
- Printed electronics / NFC manufacturers
- CMOS / VLSI fabrication partners
- Cold-chain logistics companies
- AI & cloud service providers
- Seafood Export Development Authority (MPEDA)
- Research institutes (IITs, ICAR, CSIR)

KEY ACTIVITIES

- Manufacturing of biodegradable smart films
- Fabrication of disposable sensor films
- Development of reusable NFC electronic tags
- AI model development for shelf-life prediction
- Cloud platform management
- Quality testing and certification
- R&D for new sensing technologies
- Customer support and software updates

KEY RESOURCES

- Patented SensoPack technology
- Chitosan biofilm formulation
- MIP sensors & anthocyanion indicator
- CMOS ASIC / NFC technology
- AI algorithms & cloud platform
- Intellectuul Property (IP)
- Skilled R&D and operations team
- Manufacturing facilities

VALUE PROPOSITION

- Extends shrimp shelf life by reducing microbial spoilage
- Real-time freshness monitoring
- AI-based remaining shelf-life prediction
- Smart NFC-enabled packaging
- Reduces food waste
- Supports export quality assurance
- Improves cold-chain traceability
- Biodegradable and sustainable packaging
- Reusable electronic tag minimizes e-waste

TECHNOLOGY ARCHITECTURE (4-LAYER)

1. Active Biofunctional Film (Protects & Extends Shelf Life)
2. Disposable Smart Sensor Film (Detects Spoilage Markers)
3. Reusable NFC Tag (Stores & Transmits Data)
4. AI Cloud Platform (Predicts & Provides Insights)

KEY BENEFITS

- ✓ Improved food safety
- ✓ Reduced food waste
- ✓ Enhanced export competitiveness
- ✓ Lower operational losses
- ✓ Sustainable & circular solution

CUSTOMER RELATIONSHIPS

- Long-term contracts with exporters
- Technical support and training
- AI dashboard subscription
- Cloud-based monitoring services
- Product maintenance and updates
- Dedicated enterprise support

CHANNELS

- Direct B2B sales
- Packaging distributors
- Seafood export associations
- Trade exhibitions and conferences
- Digital platform and mobile app
- Strategic partnerships

CUSTOMER SEGMENTS

- Shrimp processors and exporters
- Seafood importers & distributors
- Retail supermarkets and chains
- Cold-chain logistics providers
- Hotels, restaurants and HoReCa
- Food safety agencies and regulators
- Export certification and inspection bodies

COST STRUCTURE

- Chitosan extraction and processing
- Essential oil procurement
- Sensor fabrication (MIP & anthocyanin)
- NFC tag and CMOS ASIC manufacturing
- Cloud infrastructure and AI development
- R&D and product innovation
- Quality testing and certification
- Marketing and sales
- Logistics and supply chain
- Customer support and operations

REVENUE STREAMS

- Sale of disposable smart packaging films
- Sale / lease of reusable NFC smart tags
- AI freshness prediction subscription (SaaS)
- Cloud dashboard subscription
- Cold chain monitoring subscription
- Licensing of sensor technology
- Data analytics and insights
- Custom packaging solutions
- Annual maintenance contracts (AMC)

SOCIAL & ENVIRONMENTAL IMPACT

- Reduces food waste and economic loss
- Supports sustainable seafood export
- Biodegradable materials, eco-friendly
- Minimizes electronic waste through reusable tags
- Contributes to UN SDGs (2, 12, 13, 14, 17)

SENSOPACK™ JOURNEY – FROM PACKAGE TO AI INSIGHT

Shrimp Packed in SensoPack → Active Film Protects & Extends Shelf Life → Sensor Film Detects Spoilage Gases → Tag Stores & Transmits Data (NFC) → Data to Cloud (Secure) → AI Predicts Shelf Life & Quality → Freshness Status (SAFE / CAUTION / UNSAFE)


Smart Sensors Inside Package
Invisible Protection. Intelligent Monitoring.
TMA / TVB-N Sensor
O2 Oxygen Sensor
Temperature Sensor
NFC Chip
Detects Spoilage Gases (TMA, TVB-N)
Monitors Oxygen & Temperature
Cold Storage & Transport
Monitors Conditions in Storage & Transit
Data Sent Wirelessly
Real-Time Temperature Tracking
Cold Chain Visibility & Traceability
AI
AI Analysis
Predicts Shelf Life & Safety
Real-Time Scan by User
Scan Anytime. Know Instantly.
SensoPack
Fresh
4 Days Left
SensoPack
Smart Freshness Shrimp Pack
NFC
Smart Insights
Freshness Status & Alerts
AI Predicts Shelf Life & Sends Alerts
Still Fresh
Safe for 2 More Days
Caution
Consume Within 24 Hours
Warning
Spoiling
Eat Within 6 Hours!
SensoPack
Shrimp Pack
Pack ID: SPK2505178
92%
Freshness Score
Fresh
Safe for 2 More Days
Estimated Shelf Life
2 Days Left
Scan with Smartphone to Check Freshness
Improved Food Safety
Reduces Risk of Foodborne Illness
Reduce Food Waste
Minimize Losses Across the Supply Chain
Maximize Value
Better Quality, Better Price, Higher Profit
Sustainable Solution
Biodegradable Film, Eco-Friendly Future
SensoPack – Smart Packaging. Real-Time Freshness. Safer Seafood. Better Tomorrow.

Thank You